

Ryan Hung

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EDUCATION

Stony Brook University

Expected May 2028

B.S. in Computer Science, Applied Math

GPA: 3.8/4.0

- **Relevant Coursework:** Systems Fundamentals, Data Structures, Graph Theory, Linear Algebra
- **Ongoing Coursework:** Software Development, Analysis of Algorithms, Survey of Probability & Statistics
- **Extracurriculars:** Game Development Club, Computing Society, Robotics Club, Hall Council (Secretary)

EXPERIENCE

Cybersecurity Research Assistant

Aug 2025 – Present

Ethos Lab Group, Stony Brook University

Stony Brook, NY

- Assisted in the ongoing development of Inception, a honeypot deployment platform that uses templated configurations to deploy web honeypots with Python, Apache, and AWS for capturing automated bot traffic.
- Analyzed 8 months of JSON log data using regex-driven parsing, frequency analysis, and correlation to identify behavioral patterns and HTTP/TLS attack indicators; utilized Python libraries such as pandas and matplotlib
- Contributed to ongoing research on bot fingerprinting and detection, evaluating browser, network, and behavioral signals to improve identification of malicious automated clients.

Linear Algebra Teaching Assistant

Aug 2025 – Dec 2025

Stony Brook University

Stony Brook, NY

- Held 3 hours of weekly office hours, guiding 50+ students with homework and understanding course material
- Assisted with grading homework assignments and exams for 100+ undergraduate students, providing constructive feedback

Programming & Gaming Instructor

Oct 2024 - June 2025

TD3 Innovative Gaming

Long Island, NY

- Conducted 2 hour monthly sessions teaching 10+ K-8 students math and programming topics
- Maintained a positive attitude towards learning and motivated students to develop a passion for learning in an exciting and engaging environment through video games and other interesting mediums

PROJECTS

Mandarin Lab – React, TypeScript, Node.js, PostgreSQL, Supabase

Oct 2025 – Present

- Designed and built a full-stack application that performs structured Chinese sentence analysis, including tokenization, pinyin/zhuyin, grammatical roles, and usage metadata.
- Implemented secure authentication and persistence using Supabase OAuth and PostgreSQL with row-level security (RLS) to enforce per-user data isolation.
- Modeled complex, evolving linguistic data using PostgreSQL JSONB while supporting efficient indexed queries for saved sentence history and re-analysis.

MacOS Style Portfolio – React, TypeScript, Tailwind CSS, GSAP

Sep 2025 – Dec 2025

- Built a macOS-inspired portfolio interface featuring an interactive Apple-style Dock and windowed applications using React and Tailwind CSS.
- Implemented draggable, resizable, and focus-aware application windows with reusable Higher-Order Components and shared UI logic.
- Integrated GSAP animations to deliver smooth window transitions, dock magnification effects, and UI interactions.

Dynamic Memory Allocator – C

Sep 2025 - Dec 2025

- Implemented malloc/free/realloc using segregated freelists and block coalescing/splitting strategies.
- Handled heap alignment, corrupt pointer detection, and heap growth to ensure memory safety.

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript/TypeScript, HTML/CSS

Frameworks: React, Node.js, Express

Database: PostgreSQL, MongoDB

Developer Tools: Git, Bash, Linux, AWS, Docker

Concepts: Version Control, OOP, RESTful APIs, Unit Testing, Concurrency